

SignBridge: Product Requirements Document

Real-Time AI-Powered Bidirectional Video Communication for Sign Language Users

Document Version	Target Platform	Status	Classification
1.0 (Comprehensive)	Web (WebRTC) / Desktop	Approved for Design & Dev	Engineering & Product Spec

1. Executive Summary & Problem Statement

Over 70 million Deaf and hard-of-hearing individuals worldwide use sign language as their primary mode of communication. Standard remote video conferencing platforms (e.g., Zoom, Google Meet, Microsoft Teams) do not provide native continuous sign language translation, placing the burden of communication entirely on human interpreters or text-only live captions.

SignBridge is a real-time, browser-accessible communication bridge that translates sign language video streams into synchronized text and synthesized natural AI speech, while simultaneously converting spoken dialogue from hearing participants into live captions and sign language glosses.

2. Key User Personas

PERSONA A: DEAF / HARD OF HEARING INDIVIDUAL

Needs & Pain Points: Communicates naturally via sign language (ASL/ISL). Needs frictionless video calling without having to type everything in the chat window, with full autonomy over the synthetic voice representing them.

PERSONA B: HEARING PEER / PROFESSIONAL / COLLEAGUE

Needs & Pain Points: Has little or no understanding of sign language. Needs to hear or read seamless, low-latency translations of what the Deaf participant signs, while speaking normally through their microphone.

3. End-to-End System Architecture & Pipeline

The system operates in real time during two-way WebRTC video calls across two concurrent pipelines:

3.1 Signer → Listener Pipeline (Sign-to-Speech / Sign-to-Text)

- Video Ingestion:** Ingests webcam stream (720p/1080p at 30 fps) through client-side WebRTC canvas buffers.
- Landmark & Pose Extraction:** Employs MediaPipe Holistic to extract 543 spatial coordinates (hand joints, facial keypoints, shoulder/arm pose) per frame without transmitting raw video frames to backend servers for landmark computation.
- Temporal Continuous Sequence Model:** Spatial-temporal landmark sequences ($T \times N$ keypoints) are processed by an encoder-decoder Transformer / BiLSTM model to perform Continuous Sign Language Recognition (CSLR) and generate text glosses and full English sentences.
- Text-to-Speech (TTS) & Caption Overlay:** Sentences are dispatched to a low-latency neural TTS engine and broadcast as an audio stream to the hearing listener alongside dynamic subtitle overlays.

3.2 Listener → Signer Pipeline (Speech-to-Text / Visual Assist)

- 1. **Audio Capture & ASR:** Captures spoken microphone audio and feeds it to an Automatic Speech Recognition pipeline (Whisper/VOSK) streaming text tokens in real time.
- 2. **Captioning & NLP Parsing:** Translates rapid speech into clear sentence structures displayed directly over the video frame with high-contrast customizable fonts.

4. Functional Requirements & Specifications

Module	Feature Description	Technical Requirement	Priority
Video Calling (WebRTC)	Peer-to-peer audio/video streaming with adaptive bitrate for unstable connections.	WebRTC Mesh / SFU, STUN/TURN servers	P0 (Must Have)
Continuous Sign Detection	Real-time tracking of hand gestures, fingerspelling, and non-manual markers (facial expressions).	MediaPipe Holistic + Sequence Transformer	P0 (Must Have)
AI Voice Synthesis	Dynamic generation of speech audio for the signing user with selectable pitch/accent profiles.	Edge TTS / FastSpeech API (< 250ms audio buffer)	P0 (Must Have)
Live Subtitle Engine	Bidirectional closed captioning overlay with customizable sizing, contrast, and auto-scroll.	HTML Canvas / DOM Overlay with WebVTT stream	P0 (Must Have)
Automated Call Summary	Post-call automated generation of full transcript and structured bulleted key takeaways.	LLM Summarization Engine + MySQL archive	P1 (Important)
Vocabulary Customization	User ability to add custom domain-specific jargon, names, and shortcuts to enhance recognition.	User Dictionary Vector Store / MySQL	P2 (Nice to Have)

5. Non-Functional & Performance Requirements

- **End-to-End Latency:** The total time from physical gesture completion to listener audio output must not exceed $T_{\{latency\}} \leq 800 \text{ ms}$.
- **Accuracy Target:** Target minimum 88% Word Accuracy Rate (WAR) on standard everyday conversational sign datasets.
- **Client Performance:** Landmark tracking must execute at ≥ 25 FPS on consumer-grade hardware without requiring dedicated external GPU hardware.
- **Data Privacy & Security:** Video streams are processed ephemerally. Landmark data and transcripts are encrypted in-transit ($\text{ext}\{TLS\ 1.3\}$) and at-rest ($\text{ext}\{AES-256\}$).

6. Database Schema & Architecture Design

The backend operates with a clean relational structure supporting user accounts, custom profile voices, call session tracking, and encrypted transcript summaries:

Table Name	Primary Fields	Description
users		

Table Name	Primary Fields	Description
	id (PK), email, password_hash, default_mode, voice_profile_id, created_at	User profiles, preferences (Deaf/Hearing), and voice settings.
call_sessions	id (PK), caller_id (FK), receiver_id (FK), room_token, start_time, end_time, status	Active and historical WebRTC meeting session metadata.
call_transcripts	id (PK), session_id (FK), sender_id (FK), message_type, content, timestamp	Time-stamped caption segments for review and call minutes compilation.
voice_presets	id (PK), voice_name, gender, accent, pitch_rate, provider_key	Configured AI TTS synthetic voices available to the user.

7. Implementation Roadmap & Milestones

Phase	Timeline	Deliverables & Milestones
Phase 1: Research & Landmarks	Weeks 1 – 3	WebRTC video call baseline setup; client-side MediaPipe landmark extraction pipeline.
Phase 2: Translation Engine	Weeks 4 – 7	Model training on core sign vocabulary; gesture-to-text inference API integration.
Phase 3: TTS & Voice Sync	Weeks 8 – 10	Low-latency TTS streaming integration; customizable voice profile selection.
Phase 4: Testing & Optimization	Weeks 11 – 13	End-to-end latency optimization (<800ms), user testing with Deaf community participants.

SignBridge Vision: Creating an equitable, seamless, and natural communication environment where sign language is instantly understood by everyone on any video call.